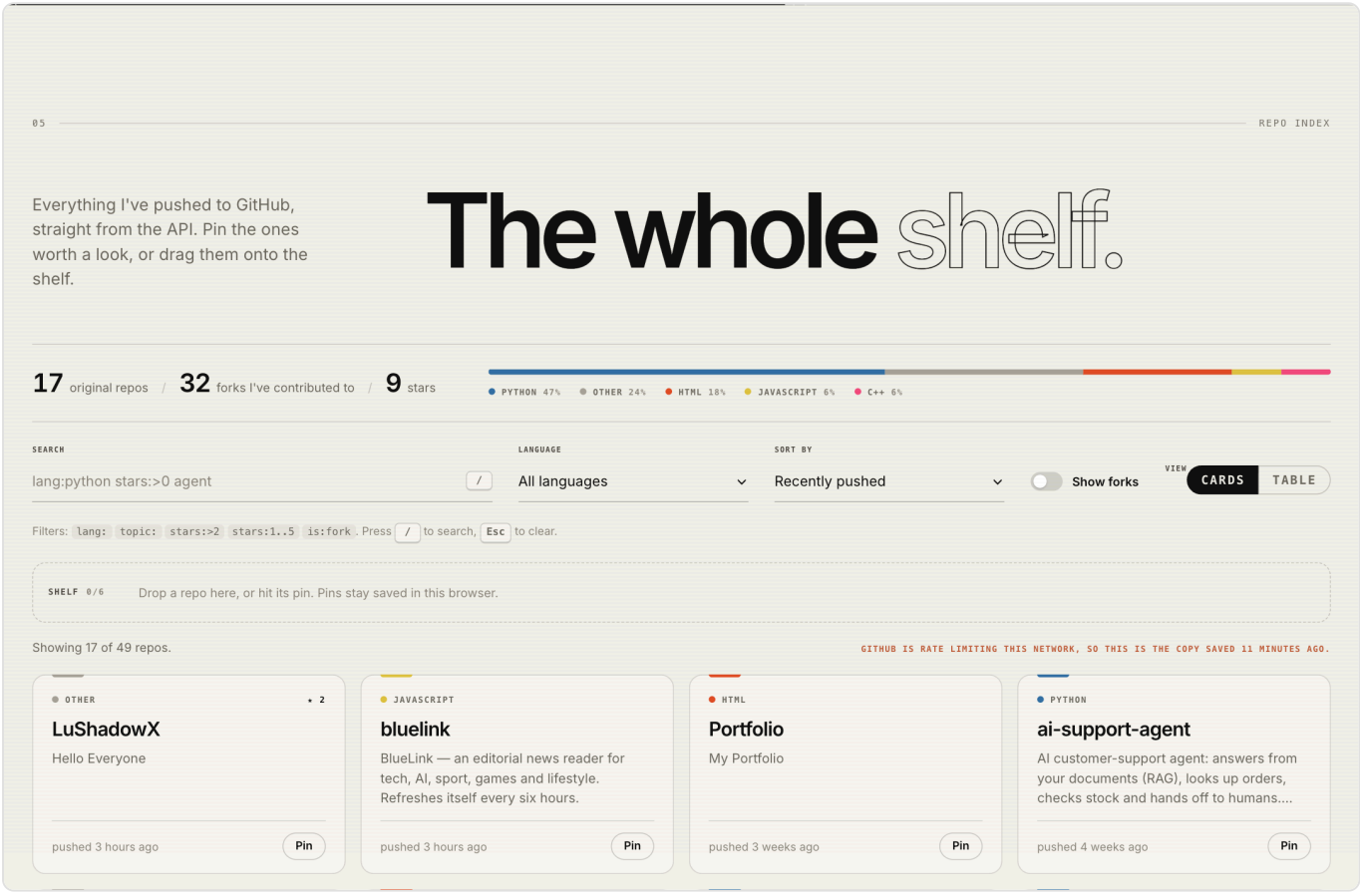
CSE326 · INTERNET PROGRAMMING LABORATORY · GRADE UPGRADATION PROJECT

LuShadowX Portfolio — a client-side web application

A responsive, accessible personal site built with plain HTML5, CSS3 and JavaScript (ES modules), deployed on GitHub Pages. It now includes a live GitHub repository index, a validated contact form, and browser API features that cover every unit of the CSE326 syllabus.



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REGISTRATION NO.	
PROGRAMME	B.Tech Computer Science and Engineering, Semester 5
COURSE	CSE326 Internet Programming Laboratory (L1 T0 P2, 2 credits)
LIVE SITE	lushadowx.github.io/CSE326-Portfolio
SOURCE CODE	github.com/LuShadowX/CSE326-Portfolio
SUBMITTED	October 2026

01

About the project

The portfolio started as a static showcase: intro animation, about section, skills, certificates and projects. For this upgrade I added features that make it a real client-side application. Each one has a reason to exist on a portfolio rather than being a demo bolted on.

- **Repo index**: every repository on my GitHub, fetched live from the GitHub REST API. It has a search box that understands filters like `lang:python stars:>2`, a card view and a sortable table view, and a "shelf" you can drag repositories onto. The shelf is saved in the browser.
- **Contact form**: rebuilt with JavaScript and regular-expression validation, accessible error messages, a character counter, a spam honeypot and a draft that survives a page reload.
- **From the desk**: live clocks in my time zone and the visitor's, an embedded OpenStreetMap of LPU, an opt-in "how far away are you?" check using the Geolocation API, and a custom audio player for a short voice note.
- **Engineering**: SEO metadata and structured data, a skip link and keyboard support, an offline fallback for the API, a debug mode, and code split into ES modules.

Technology

Markup	HTML5: semantic sections, forms, table, SVG, video, audio, iframe, template
Styling	CSS3: custom properties, Flexbox, Grid, media and container queries, `:has()`, pseudo-elements, counters, keyframe animations, transitions
Scripting	JavaScript (ES2022 modules): no framework, no build step
Browser APIs	DOM, Fetch, Web Storage (local + session), Geolocation, Drag and Drop, HTMLMediaElement, IntersectionObserver, Intl, Performance
External data	GitHub REST API (`/users/LuShadowX/repos`), GitHub contributions API, OpenStreetMap embed
Tooling	VS Code, Chrome DevTools (Elements, Console, Network, Application, device mode), Git and GitHub, GitHub Pages, GitHub Copilot

02

Course outcomes covered

CO1
Internet technologies and HTML5 for web page development

CO2
Semantic HTML, web accessibility and web forms

CO3
CSS for responsive, visually appealing interfaces

CO4
JavaScript for dynamic client-side applications

CO5
Interactive apps with DOM, events and browser APIs

CO6
Debugging, analysing and deploying with GitHub Pages

CO	WHERE IT SHOWS UP IN THE PROJECT	FILES
CO1	Client–server model in practice: the browser requests JSON from GitHub's server and static files from GitHub Pages. HTML5 document structure with full head metadata. Text formatting, lists, links, images, inline SVG (icons, heatmap, language bar generated at runtime), `<video>` menu previews, `<audio>` voice note, OpenStreetMap `<iframe>`. SEO: description, canonical, Open Graph, JSON-LD, sitemap.xml, robots.txt.	`index.html`, `sitemap.xml`, `robots.txt`
CO2	`header`, `nav`, `main`, `section`, `article`, `aside`, `footer`, `figure`, `time`, `dl`. Skip link, ARIA live regions, `aria-invalid` / `aria-describedby` on form errors, `aria-sort` on the table, `aria-pressed` on pin buttons, keyboard alternative for drag and drop. Forms with search, email, text, select, checkbox, radio (`fieldset` / `legend`), range, textarea. Data table with `caption`, `thead`, `scope`.	`index.html`
CO3	Design tokens in CSS variables. Grid for the card list and controls, Flexbox for chips and rows. Media queries at 980, 760 and 480 px, plus a container query that turns the table into stacked rows. Pseudo-classes (`:hover`, `:focus-visible`, `:checked`, `:invalid`, `:has()`, `:empty`), pseudo-elements (`::before`, `::after`, `::placeholder`, slider thumbs), attribute selectors (`[data-fork="true"]`, `[aria-sort]`, `[hidden]`), CSS counters, custom-styled form controls, keyframe animations and transitions, `prefers-reduced-motion`.	`assets/css/repo-index.css`, `assets/css/main.css`
CO4	`const` / `let`, all primitive types, `??` and `?.`, template literals, string methods, `Math` and `Intl.NumberFormat`, `switch` and ternaries, normal, arrow and `async` functions, closures (`debounce`, `cache`), arrays (`map`, `filter`, `reduce`, `sort`, `some`, `every`, `splice`), objects with destructuring and spread, `Map` and `Set`.	`assets/js/modules/*.js`
CO5	DOM selection and creation (`<template>` cloning, `insertRow`, `DocumentFragment`). Click, input, change, keydown, focusout, submit, drag and media events. Regex search qualifiers and form validation, date and time zones with `Intl`, localStorage and sessionStorage, Fetch + JSON with timeout and fallback, Geolocation, HTML Drag and Drop, audio events driving a custom player.	`repo-index.js`, `contact-form.js`, `desk.js`
CO6	`?debug` mode with `console.table` and `performance.mark/measure`. Global error and unhandled-rejection listeners. A `guard()` wrapper so one failing feature can't break the page. `try/catch/finally` with custom error data. Code split into ES modules, lazy API loading with IntersectionObserver, debounced input, preconnect. Hosted on GitHub Pages from the repository.	`app.js`, `modules/debug.js`, `modules/storage.js`

03

Syllabus coverage, unit by unit

Unit I: HTML Fundamentals

TOPIC	IMPLEMENTATION
Internet and web technologies, client–server	Static site served by GitHub Pages. The browser (client) calls the GitHub REST API (server) and handles HTTP status codes, including `403` rate limiting with a fallback.
HTML5 document structure, elements and attributes	`<!DOCTYPE html>`, `lang`, charset and viewport meta, data-* attributes used as JS hooks
Text formatting, lists, hyperlinks, images	`strong`, `em`, `small`, `kbd`, `code`. Ordered list for the pinned shelf, unordered lists for cards and topics. External links with `rel="noopener"`, lazy-loaded images with `alt`.
SVG graphics	Inline SVG icons, the contribution heatmap, and a language-share bar generated from data at runtime
Audio, video, iframes	Voice-note `<audio>` with two `<source>` formats, menu preview `<video>`, OpenStreetMap `<iframe>` with `title` and `loading="lazy"`
SEO fundamentals, best practices	Meta description, canonical link, Open Graph and Twitter tags, JSON-LD `Person` schema, sitemap.xml, robots.txt, a single `h1` per view, and a `noscript` message

Unit II: Semantic HTML and Forms

TOPIC	IMPLEMENTATION
Semantic elements, page structure	Each section uses `section` + `aria-labelledby`. Cards are `article` elements with a `footer` and a `time datetime`. The desk card is an `aside`.
Accessibility fundamentals and standards (WCAG)	Skip link, focus-visible outlines, live regions, `aria-pressed`, `aria-sort`, `aria-invalid`, labelled controls, keyboard reordering of pins (arrow keys / Delete), reduced-motion support
Forms, controls, input types and attributes	search, email, text, select, checkbox, radio group in `fieldset`, range, textarea. Attributes include `required`, `minlength`, `maxlength`, `autocomplete`, `inputmode`, `placeholder` and `novalidate`.
Form validation basics	HTML constraints are in the markup. JavaScript then takes over for friendlier, accessible messages.
Tables and data representation	Repository table with `caption`, `thead` / `tbody`, `th scope="col"` and `scope="row"`, and sortable headers
Responsive content structuring	Content order works on a single column. Controls reflow from 5 columns to 2 to 1.

Unit III: Cascading Style Sheets

TOPIC	IMPLEMENTATION
Selectors and specificity	Class-based BEM-style names (`.repo-card__name`), low specificity, state classes (`.is-over`, `.is-dragging`)
Colours, backgrounds, units, typography	Hex with alpha, gradients and an inline SVG data-URI chevron. `rem`, `px`, `%`, `vw` and `clamp()` for fluid type. Inter and monospace pairing, `tabular-nums`.
Box model, positioning	`box-sizing`, padding and border on cards. Absolute positioning for the drop marker, `kbd` hint and map caption. Fixed skip link.
Flexbox, Grid	Grid: card list (`auto-fill` + `minmax`), controls, overview. Flexbox: chips, card footer, status line.
Pseudo-classes, pseudo-elements, attribute selectors	`:has()`, `:checked`, `:invalid`, `:empty`, `:focus-visible`, `:placeholder-shown`. `::before` / `::after` for dots, counters and sort arrows. `[data-fork="true"]`, `[aria-pressed="true"]`, `[data-label]`.
Form styling	Custom switch checkbox, segmented radio buttons, custom select arrow, styled range slider, error states
CSS variables, responsive design	Tokens per section (`--ink`, `--paper`, `--accent`), per-item variables (`--lang`, `--progress`), media queries and a container query
Animations and transitions	Loading shimmer, "shelf full" shake keyframes, card lift and underline transitions, switch knob, all disabled under `prefers-reduced-motion`

Unit IV: JavaScript Fundamentals

TOPIC	IMPLEMENTATION
Variables, constants, data types	`const` by default, `let` only where values change. Strings, numbers, booleans, `null`, objects, arrays, `Date`, `Map`, `Set`.
Operators and expressions	Nullish coalescing, optional chaining, logical short-circuit, exponent (`**` in the haversine formula), spread
Strings, numbers, booleans	`replace`, `split`, `includes`, `padStart`, `toLowerCase`. `Math.round`, `toFixed`, `Number.isInteger`, `Intl.NumberFormat('en-IN')`.
Conditionals	`if/else`, `switch` for qualifiers and key handling, ternaries
Functions and scope	Function declarations, arrow functions, async functions, default parameters, rest arguments, block scope, module scope
Arrays and objects	`map` / `filter` / `reduce` / `sort` / `find` / `some` / `every`, `Object.fromEntries`, `Object.entries`, destructuring, the comparator lookup table `SORTS`

Unit V: Interactive Web Development

TOPIC	IMPLEMENTATION
Loops, higher-order functions, callbacks	`for...of` in `timeAgo`, a filter → filter → filter → sort pipeline, `debounce` returning a function, a callback-based Geolocation API wrapped in a Promise
DOM selection and manipulation	`querySelector` helpers, `<template>` cloning, `replaceChildren`, `classList`, `dataset`, `setAttribute`, `insertRow` / `insertCell`

TOPIC	IMPLEMENTATION
Event handling, dynamic content	Event delegation on lists. `click`, `input`, `change`, `keydown`, `focusout`, `submit`, `dragstart` / `dragover` / `drop` / `dragend`. The `/` shortcut.
JS accessibility practices	Focus restored after re-render, focus moved to the first invalid field, live-region announcements
Form validation, regular expressions	`/^\p{L}[\p{L}\p{M} .'-]{1,59}$/u` for names, an email pattern, link counting, the search-qualifier parser `/\b(lang|language|topic|stars|is):("[^"]*"|\S+)/gi`, and the time-zone offset parser
Date and time	Live clocks with `Intl.DateTimeFormat` across time zones, `Intl.RelativeTimeFormat` ("pushed 3 days ago"), a ticker aligned to whole seconds
Audio and video events	`loadedmetadata`, `timeupdate`, `play`, `pause`, `ended`, `error` drive the voice-note player. Video `playing` / `waiting` drive the menu previews.
Web Storage	localStorage: pins, view preferences, contact draft. sessionStorage: 30-minute API cache with TTL. Every access is wrapped because storage can throw.
Fetch API and JSON	`fetch` + `await res.json()`, `res.ok` check, `AbortController` timeout, fallback to a local JSON snapshot
Geolocation API	Asked only when the visitor clicks. Every error code is handled. Distance is computed locally with the haversine formula and never stored.
Drag and Drop API	Cards drag onto the shelf, chips reorder by drop position, and dragging a chip back to the list unpins it

Unit VI: Web Application Development and Deployment

TOPIC	IMPLEMENTATION
Debugging, browser developer tools	`?debug` flag: `console.table` of repos and load timings through the Performance API. Tested with the Network panel (offline / throttled), Application panel (storage) and device mode.
Error handling	try/catch/finally, error objects carrying `rateLimited`, a three-step fallback (cache → API → snapshot), a visible retry button, global error listeners, `guard()` per feature
Interactive UI, client-side application design	A single `state` object, pure helpers (`parseQuery`, `normalize`), and render functions that rebuild the UI from state
Code organization	ES modules: `app.js` entry, feature modules, and shared `utils`, `storage` and `debug`
Website performance	API call deferred until the section is near (IntersectionObserver), debounced search, `DocumentFragment` rendering, preconnect, lazy iframe and images, session cache
GitHub repository management, GitHub Pages	Source on GitHub, deployed from the repository root with GitHub Pages. README documents setup, structure and debugging.
Improving web apps with GitHub Copilot	Used AI suggestions (GitHub Copilot) to review the code, catch edge cases and refactor. Every change was then tested by hand in DevTools.

04

List of practicals: where each one lives

#	PRACTICAL	COVERED BY	
1	Registration form with HTML input controls, CSS selectors, box model, focus styles	Contact form: input types, focus and error states, padded controls	✓
2	Page with text formatting, lists, links, images, SVG, audio, video, iframe	index.html: all present, including the voice note, menu videos and map iframe	✓
3	Responsive page: Flexbox nav, Grid content, media queries	Top bar (Grid/Flex), card grid, breakpoints and container query	✓
4	CSS variables, pseudo-classes and elements, transitions, animations	repo-index.css: tokens, `:has()`, counters, shimmer, shake, hover lift	✓
5	JS variables, constants, types, operators, control flow	All modules. `parseQuery` and `moodAt` are good walk-throughs.	✓
6	Normal and arrow functions, arrays and objects	`distanceKm`, `SORTS`, `normalize`, `visibleRepos`	✓
7	Click, keyboard and form-change events updating the page	Pin buttons, `/` and Esc keys, arrow-key reordering, filters	✓
8	Form validation with JS and regular expressions	`contact-form.js` RULES object	✓
9	Execution context, scope, closures, loops, callbacks, higher-order functions	`debounce`, `cache`, `wrap`, `locate`, array pipelines	✓
10	HTML5 page with semantic elements and basic CSS layout and typography	The whole site	✓

05

Testing

CHECK	RESULT
Live API	Repos load and the source label reads "Live from the GitHub API"
Rate limit / offline	A 403 from GitHub falls back to the snapshot and the page says so. With both unavailable, an error message and "try again" button appear.
Search	`lang:python stars:>0` → 3 results. `stars:abc` → shown as unreadable with a hint.
Pins	Pin by button and by drag, reorder with arrow keys, survive a reload
Table	Header click sorts and sets `aria-sort`. Rows stack on narrow screens.
Form	An empty submit shows 4 messages and focuses the first field. The draft is restored after a reload.
Desk	London visitor: "I'm 4.5 hours ahead of you". Delhi location: ~328 km from Phagwara.
Responsive	390 px wide: no horizontal scroll, controls in one column

06

Screenshots

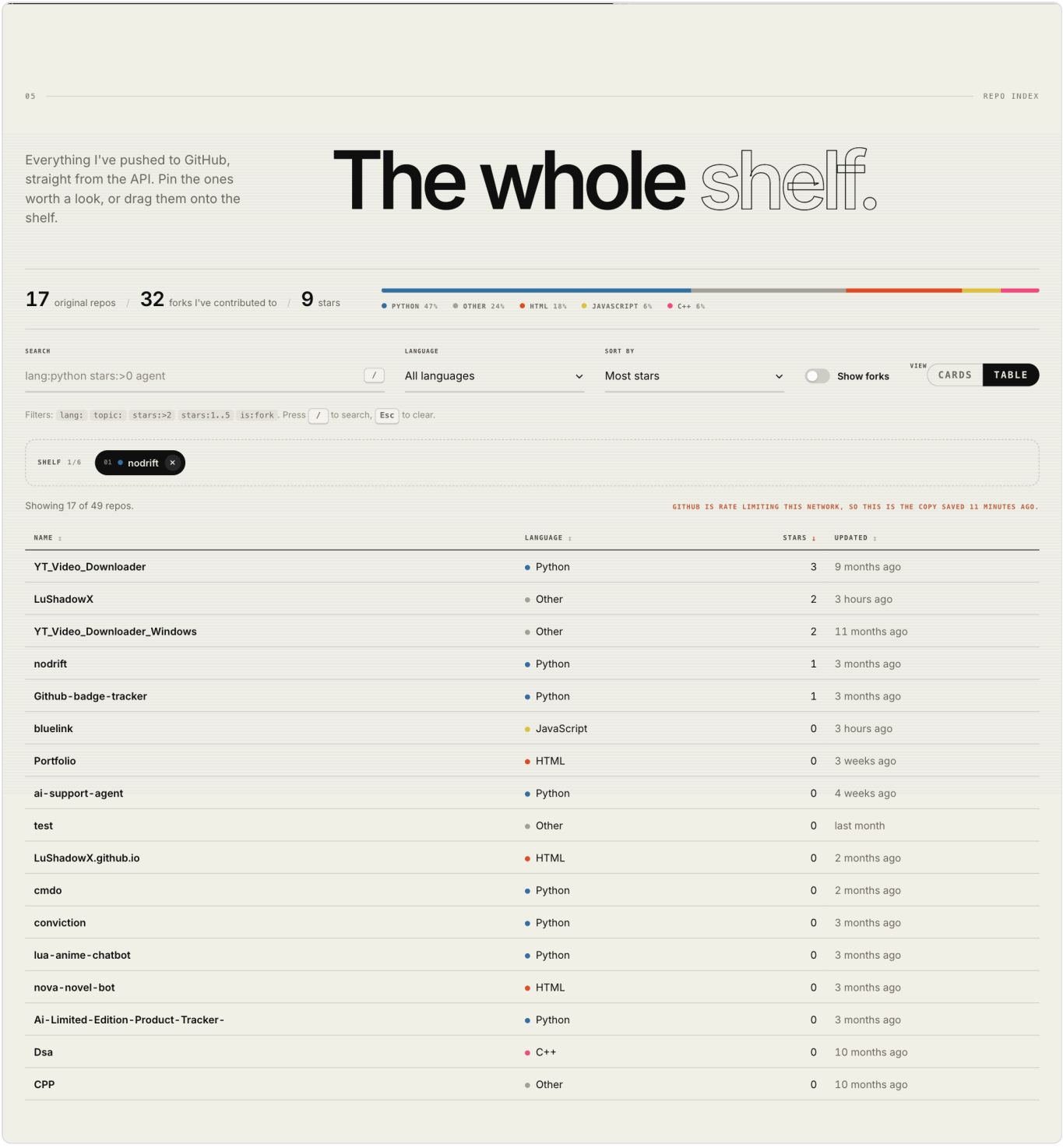
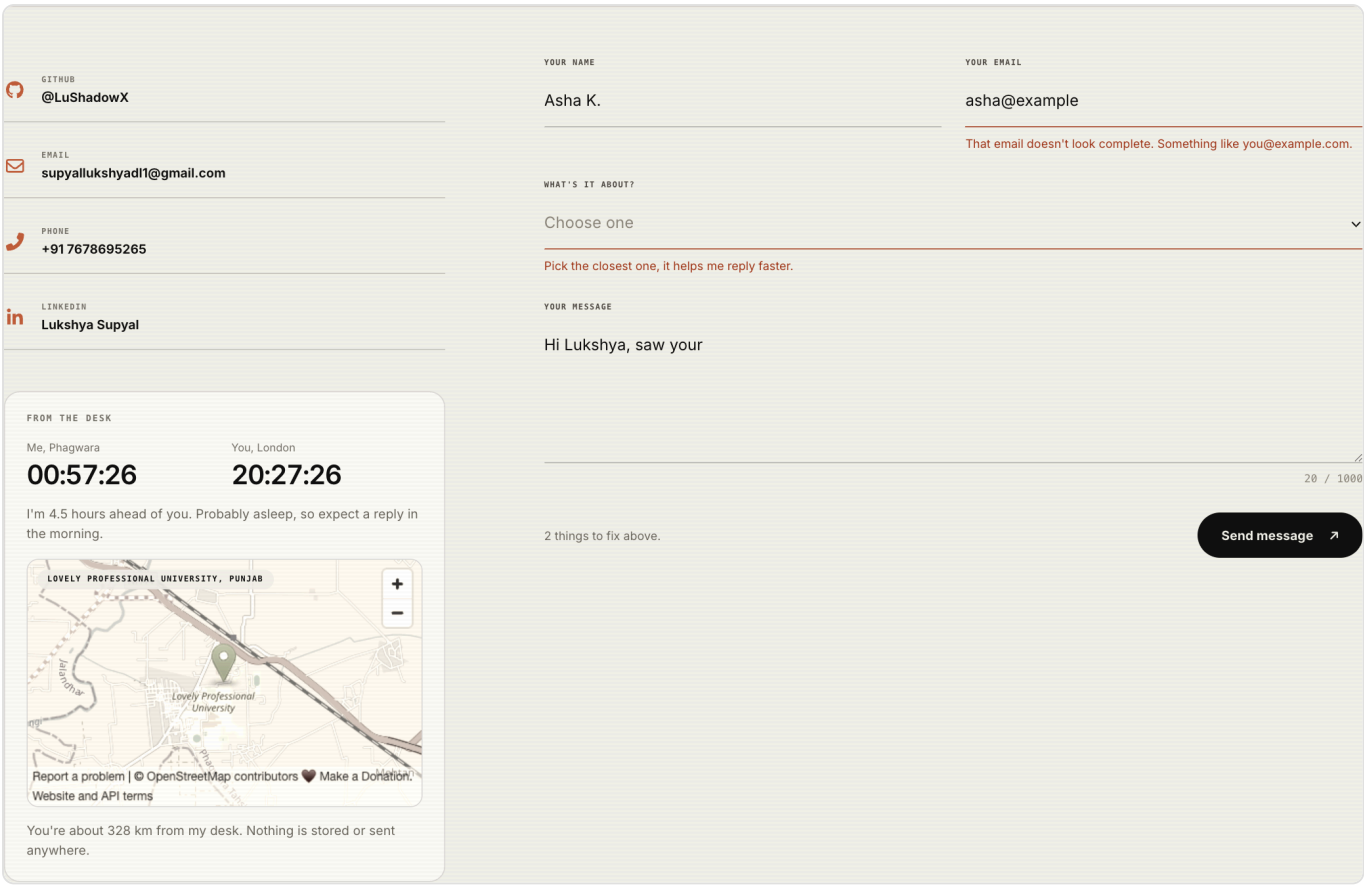


Table view sorted by stars, with two repositories pinned to the shelf.



Contact form after an invalid submit, and the "From the desk" card with clocks, map and distance.

05 — REPO INDEX

Everything I've pushed to GitHub, straight from the API. Pin the ones worth a look, or drag them onto the shelf.

The whole shelf.

17 original repos

32 forks I've contributed to

9 stars

PYTHON 47% OTHER 24% HTML 18%
JAVASCRIPT 6% C++ 6%

SEARCH

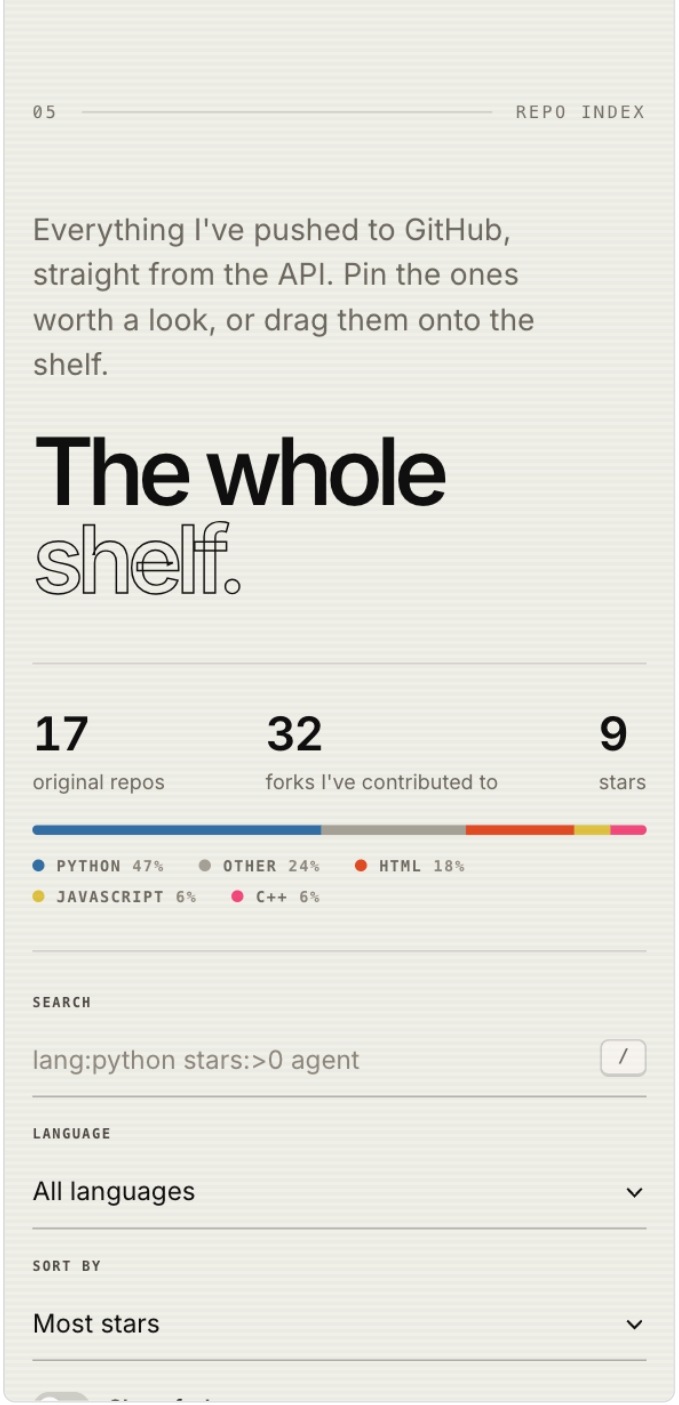
lang:python stars:>0 agent /

LANGUAGE

All languages

SORT BY

Most stars


Repo index at phone width (390 px).

SHELF 1/6 01 nodrift ×

Showing 17 of 49 repos.

GITHUB IS RATE LIMITING THIS NETWORK, SO THIS IS THE COPY SAVED 11 MINUTES AGO.

PYTHON ★ 3

YT_Video_Downloader

Download Your Favourite Youtube Video For Free

pushed 9 months ago Pin

OTHER ★ 2

LuShadowX

Hello Everyone

pushed 3 hours ago Pin

OTHER ★ 2

YT_Video_Downloader_Windows

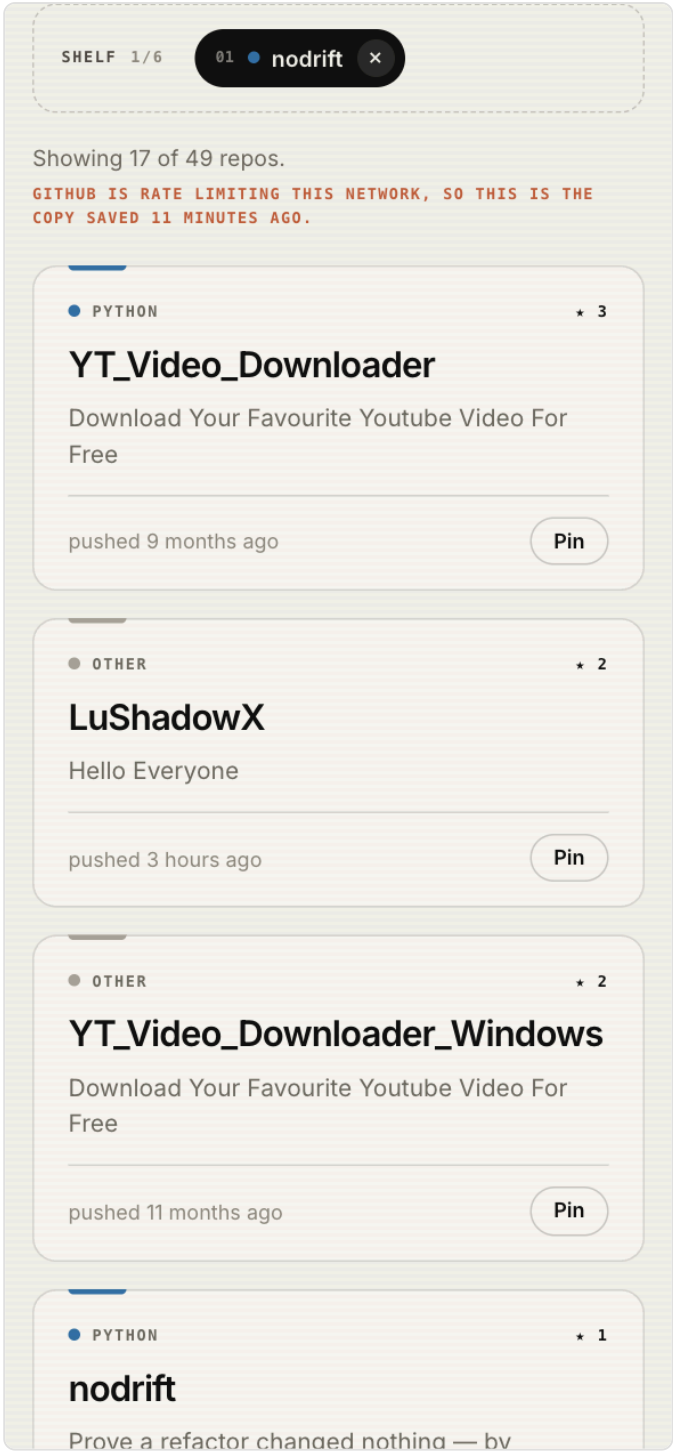
Download Your Favourite Youtube Video For Free

pushed 11 months ago Pin

PYTHON ★ 1

nodrift

Prove a refactor changed nothing — by


Cards and shelf on mobile.